

# Q1

Assume that a computer system uses demand paging for virtual memory management. A process is allocated four page frames and the execution of this process requires reference to pages 1, 2, 3, 4, and 5. Least Recent Used (LRU) algorithm is used for page replacement. Initially, there is a page loaded in each frame. Given the following string of paged out pages (where N means there is no page being paged out): 3, 1, 2, 4, 5, N, 1,

- a) what are the pages initially loaded in the frames?
- b) what is the corresponding page reference string?

# Q1 (answer)

a) Pages that are initially loaded in the memory: 3, 1, 2, 4

b) Page reference string: 5, 3, 1, 2, 4, 3, 5

# Q1 (explanation for b)

| Stack | 5 | 3 | 1 | 2 | 4 | 3 | 5 |
|-------|---|---|---|---|---|---|---|
| 4     | 5 | 3 | 1 | 2 | 4 | 3 | 5 |
| 2     | 4 | 5 | 3 | 1 | 2 | 4 | 3 |
| 1     | 2 | 4 | 5 | 3 | 1 | 2 | 4 |
| 3     | 1 | 2 | 4 | 5 | 3 | 1 | 2 |
|       | 3 | 1 | 2 | 4 | 5 | N | 1 |

1. Find the LRU order for the pages in the memory based on the page fault sequence: [5, 4, 1, 2] and arrange them in the stack according to LRU order
2. Figure out reference string based on:
  - Referencing a page that is not in memory generates a page fault, and
  - Stack implementation of LRU
3. Determine which page is referenced at step 6
  1. After step 5, the stack contains [4, 2, 1, 3] (page 3 is at bottom)
  2. Page 1 is paged out at step 7, hence it must be the LRU page at bottom of stack
  3. So, the only possibility is that page 3 is referenced at step 6 so that it's put to the top of the stack and page 1 moves to the bottom of the stack (the LRU page).

# Q2

Suppose that a process is allocated four page frames in a demand-paged virtual memory system. The time of loading and the time of last access are as shown in the table below.

| Frame # | Page # | Time Loaded | Time Last Accessed |
|---------|--------|-------------|--------------------|
| 0       | 1      | 60          | 161                |
| 1       | 0      | 130         | 160                |
| 2       | 3      | 26          | 162                |
| 3       | 2      | 20          | 163                |

Given the above memory state before the fault and the remaining page reference string of the process: 4, 0, 0, 0, 2, 4, 2, 1, 0, 3, 2, answer the following two questions.

- Calculate how many page faults will be generated if Least-Recently-Used (LRU) algorithm is used. Show your workings.
- Assuming that the size of the working set window is four, calculate the number of page faults that would occur if only the pages in the working set were loaded into the memory instead of using a fixed allocation of four frames. Show your workings.

# Q2 (a)

| Ref String |   | 4 | 0 | 2 | 4 | 2 | 1 | 0 | 3 | 2 |
|------------|---|---|---|---|---|---|---|---|---|---|
|            | 2 | 4 | 0 | 2 | 4 | 2 | 1 | 0 | 3 | 2 |
|            | 3 | 2 | 4 | 0 | 2 | 4 | 2 | 1 | 0 | 3 |
|            | 1 | 3 | 2 | 4 | 0 | 0 | 4 | 2 | 1 | 0 |
| LRU Page   | 0 | 1 | 3 | 3 | 3 | 3 | 0 | 4 | 2 | 1 |
| Out Page   |   | 0 | 1 |   |   |   | 3 |   | 4 |   |

Number of page faults: 4

# Q2 (b)

| Frame # | Page # | Time Loaded | Time Last Accessed |
|---------|--------|-------------|--------------------|
| 0       | 1      | 60          | 161                |
| 1       | 0      | 130         | 160                |
| 2       | 3      | 26          | 162                |
| 3       | 2      | 20          | 163                |

| Ref String | Working Set |   |   |   |  | Page Fault |
|------------|-------------|---|---|---|--|------------|
|            | 0           | 1 | 3 | 2 |  |            |
| 4          | 1           | 3 | 2 | 4 |  | P          |
| 0          | 3           | 2 | 4 | 0 |  | P          |
| 0          | 2           | 4 | 0 |   |  |            |
| 0          | 4           | 0 |   |   |  |            |
| 2          | 0           | 2 |   |   |  | P          |
| 4          | 0           | 2 | 4 |   |  | P          |
| 2          | 0           | 2 | 4 |   |  |            |
| 1          | 2           | 4 | 1 |   |  | P          |
| 0          | 4           | 2 | 1 | 0 |  | P          |
| 3          | 2           | 1 | 0 | 3 |  | P          |
| 2          | 1           | 0 | 3 | 2 |  |            |

Number of page faults: 7

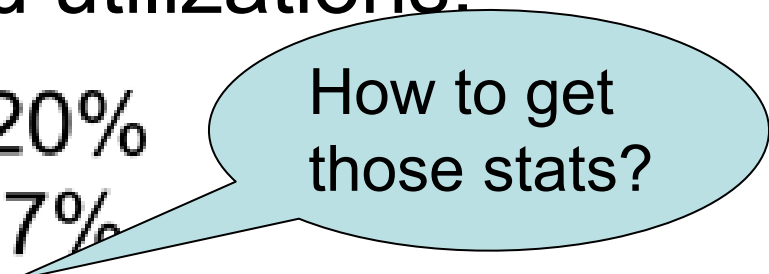
# Q3

3. Consider a demand-paging system with the following time-measured utilizations:

CPU utilization 20%

Paging disk 97.7%

Other I/O devices 5%



How to get those stats?

Which (if any) of the following will (probably) improve CPU utilization?

# Q3: improve CPU utilization?

|                      |
|----------------------|
| CPU utilization 20%  |
| Paging disk 97.7%    |
| Other I/O devices 5% |

- a. Install a faster CPU. **No.**
- b. Install a bigger paging disk. **No.**
- c. Increase the degree of multiprogramming. **No.**
- d. Decrease the degree of multiprogramming. **Yes.**
- e. Install more main memory. **Yes.**
- f. Install a faster hard disk or multiple hard disks. **Probably yes.**
- g. Increase the page size. **Depends on memory access pattern.**



# Q3

- (f) A possible improvement, for as the disk bottleneck is removed by faster response and more throughput to the disks, the CPU will get more data more quickly.
- (g)
  - Increasing the page size will result in fewer page faults if data is being accessed sequentially
  - If data access is more or less random, more paging actions could occur because fewer pages can be kept in memory and more data is transferred per page fault.